Derivative formulas through geometry

The power rule, and why it is true.

THE IDEA

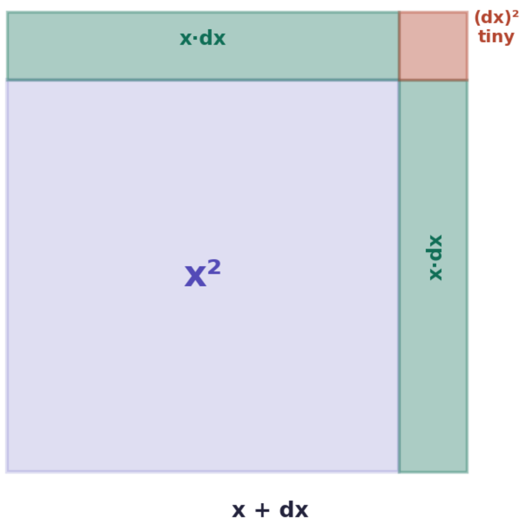
The standard derivative formulas are not arbitrary — each one can be seen geometrically. The most important is the **power rule**.

> **Power rule:** $\frac{d}{dx}(x^n) = n \cdot x^{n-1}$ — bring the power down in front, then subtract one from it.

WHY X^2 GIVES 2X

Picture a square of side x, area x^2. Grow the side by a tiny dx. Two new strips appear, each of area $x \cdot dx$, plus a tiny corner square $(dx)^2$ so small it can be ignored. So the growth is $2 \cdot x \cdot dx$; divide by dx and you get **2x**.

why $d(x^2)/dx = 2x$



growth = two strips = 2·x·dx → divide by dx → 2x

THE RULES YOU NEED

Rule	Formula
Power	$\frac{d}{dx}(x^n) = n \cdot x^{n-1}$
Constant	$\frac{d}{dx}(c) = 0$
Constant multiple	$\frac{d}{dx}(c \cdot f) = c \cdot f'$
Sum	$\frac{d}{dx}(f + g) = f' + g'$
Trig	$\frac{d}{dx}(\sin x) = \cos x$ · $\frac{d}{dx}(\cos x) = -\sin x$

WORKED EXAMPLES

CHAPTER 3 — WORKED EXAMPLES

the formulas you will use every day

THE POWER RULE

$$\frac{d}{dx}(x^n) = n \cdot x^{n-1}$$

bring the power down in front, then subtract 1 from it

EXAMPLES

$\frac{d}{dx}(x^3) = 3x^2$

$\frac{d}{dx}(x^5) = 5x^4$

$\frac{d}{dx}(x) = 1$

$\frac{d}{dx}(7) = 0$ **(a constant never changes)**

CONSTANT MULTIPLE & SUM

$\frac{d}{dx}(5x^3) = 5 \cdot 3x^2 = 15x^2$

$\frac{d}{dx}(x^3 + x^2) = 3x^2 + 2x$

differentiate each term separately, then add

WORKED — $f(x) = 4x^3 + 2x^2 - 5x + 9$

$4x^3 \to 4 \cdot 3x^2 = 12x^2$

$2x^2 \to 2 \cdot 2x = 4x$

$-5x \to -5$

$9 \to 0$

$$\frac{df}{dx} = 12x^2 + 4x - 5$$

FORMULAS FROM THIS CHAPTER

Power rule	$\frac{d}{dx}(x^n) = n \cdot x^{n-1}$
Constant	$\frac{d}{dx}(c) = 0$
Constant multiple	$\frac{d}{dx}(c \cdot f) = c \cdot f'$
Sum	$\frac{d}{dx}(f+g) = f' + g'$
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TEST — Chapter 3

1. Differentiate $f(x) = x^4$. **→ $4x^3$**
2. Differentiate $f(x) = 3x^2 + 5x - 7$. **→ $6x + 5$**
3. Differentiate $f(x) = 2x^3 - x^2 + 4x + 10$. **→ $6x^2 - 2x + 4$**